THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ratio 18 : 24 in its simplest form is: 1

(a) 4 : 4 **(b)** 4 : 3 **(c)** 3 : 4 **(d)** 3 : 5

2. The ratio 54 : 72 in its simplest form is: 1

(a) 4 : 3 **(b)** 3 : 4 **(c)** 4 : 4 **(d)** 3 : 5

3. If 2 : 7 = 10 : x, then x equals: 1

(a) 35 **(b)** 36 **(c)** 38 **(d)** 34

4. ₹ 720 is divided between A and B in the ratio 5 : 4. A's share is: 1

(a) ₹ 320 **(b)** ₹ 360 **(c)** ₹ 480 **(d)** ₹ 400

5. 2.5 km in metres is: 1

(a) 25 m **(b)** 2500 m **(c)** 250 m **(d)** 25000 m

6. 15 cm in metres is: 1

(a) 1.5 m **(b)** 0.15 m **(c)** 15 m **(d)** 0.015 m

7. If 4 : 7 = x : 21, then x is: 1

(a) 9 **(b)** 15 **(c)** 12 **(d)** 28

8. A car covers 240 km in 4 hours. In 7 hours at the same speed it covers: 1

(a) 480 km **(b)** 420 km **(c)** 280 km **(d)** 360 km

9. Rohan is 12 years old and his father is 42. The ratio of their ages is: 1

(a) 3 : 7 **(b)** 2 : 7 **(c)** 1 : 3 **(d)** 7 : 2

10. A map scale 1 cm = 5 km. A distance of 3.5 cm on the map represents: 1

(a) 1.75 km **(b)** 35 km **(c)** 17.5 km **(d)** 15 km

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a ratio? Write the ratio of 24 to 36 in simplest form. 2

12. Write the ratio of 75 paise to ₹ 3 in simplest form. 2

13. Are 4 : 9 and 12 : 27 in proportion? Why? 2

14. Divide ₹ 1200 between two friends in the ratio 3 : 5. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Divide ₹ 1500 in the ratio 2:3. 3

16. Divide ₹ 2400 in the ratio 3:5. 3

17. Divide ₹ 4200 in the ratio 4:3. 3

OR

Divide ₹ 1800 in the ratio 1:2:3. 3

18. Divide ₹ 3300 in the ratio 2:3:6. 3

OR

Divide ₹ 2800 in the ratio 3:4:7. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Define ratio and proportion. 5

(ii) Find x if $x : 12 = 15 : 20$.

(iii) Check if 6, 9, 10, 15 are in proportion.

OR

(i) Divide ₹ 7500 among A, B, C in the ratio 2 : 3 : 5. 5

(ii) If C gives ₹ 500 to A, find the new ratio.

20. **Case study:** A recipe for 6 people uses 450 g flour, 300 g sugar and 150 g butter. 5

(i) Write the ratio of flour : sugar : butter.

(ii) Find quantities for 10 people.

(iii) How many people can be served with 1.5 kg flour?

OR

(i) A map scale is 1 cm = 25 km. Find the actual distance for 7.4 cm. 5

(ii) Express the scale as a ratio.

END OF QUESTION PAPER — Total: 40 Marks

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